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Cordycepin exhibits anti-fatigue effect via activating TIGAR/SIRT1/PGC-1 α signaling pathway

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Highlights

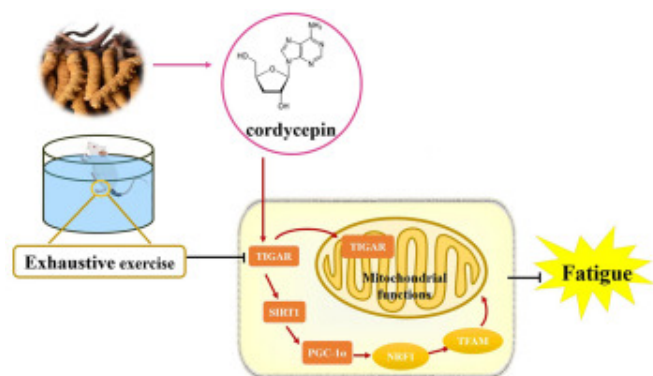
- Cordycepin exhibits anti-fatigue effect in weight-loaded swimming mice.
- Cordycepin relieves fatigue via regulating energy metabolism and pentose phosphate pathway.
- Cordycepin regulates TIGAR/SIRT1/PGC-1 α signaling pathway in weight-loaded swimming mice.

Abstract

Fatigue, a most commonly sub-health condition, may cause people more susceptible to many diseases. Cordycepin, a principal active ingredient from Cordyceps militaris, exerts various pharmacological activities including anti-diabetes, anti-inflammatory, immunomodulatory and

antioxidant effects. However, the anti-fatigue effect of cordycepin and specific mechanism remained unclear. This study aimed to investigate the beneficial effect of cordycepin on physical fatigue and elucidate the potential mechanism. 20 mg/kg, 40 mg/kg of cordycepin and 500 mg/kg taurine were respectively treated to mice for 28 days before weight-loaded swimming test. The results revealed that cordycepin significantly prolonged the weight-loaded swimming time of mice. Meanwhile, cordycepin decreased the levels of lactic acid, blood uric nitrogen, and malondialdehyde, and increased the contents of superoxide dismutase, glutathione, nicotinamide adenine dinucleotide phosphate, hepatic glycogen, muscle glycogen and ATP. The metabolomic study by GC-MS showed that eight biomarkers were found in livers, including L-lactic acid, L-asparagine, 3-phosphoglyceric acid, inosine, D-galactose, L-tyrosine, glyceric acid and L-threonine. There were seven biomarkers in gastrocnemius, including D-ribose-5-phosphate, acetic acid, propionic acid, butyric acid, palmitic acid, oxaloacetic acid and citric acid. The results of metabolomics indicated that cordycepin might relieve fatigue by regulating energy metabolism and pentose phosphate pathway. Furthermore, we found cordycepin significantly enhanced the protein levels of TIGAR, SIRT1, PGC-1 α , NRF1 and TFAM in gastrocnemius of weight-loaded swimming mice. Taken together, the present study demonstrated that cordycepin possessed an anti-fatigue effect via activating TIGAR/SIRT1/PGC-1 α signaling pathway. Our study indicated that cordycepin may be a potentially efficient candidate for fatigue.

Graphical abstract



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Introduction

Fatigue is a sub-health condition, which has caused great trouble to people's normal life due to its characteristics of persistence, unpredictability and relapse [1]. Under the fast pace of modern life, people are more likely to develop many unhealthy habits, like insufficient sleep, irregular diet, lack of exercise and excessive anxiety [2]. These unhealthy habits will induce occurrence of fatigue.

People with prolonged and excessive fatigue are more susceptible in suffering aging, depression, cancer, multiple sclerosis and Parkinson's disease [3]. At present, there are some strategies to relieve fatigue such as dietary supplements, such as taurine [4], creatine [5], L-carnitine [6], ginseng [7], and rhodiola [8]. However, the applications are limited due to inadequate mitigative effect, side-effects and high expenses [9,10]. Therefore, it is indispensable to find more effective candidate with fewer side effects to alleviate fatigue.

TP53-induced regulator of glycolysis and apoptosis (TIGAR) plays an important role in maintaining homeostasis of cellular energy metabolism [[11], [12], [13]]. The previous study has documented that TIGAR protects ischemic brain injury via enhancing the expression of glucose-6phosphate dehydrogenase (G6PD) which is the pentose phosphate pathway (PPP) rate-limiting enzyme [14]. Thus, TIGAR can inhibit glycolysis and promote glucose metabolism to PPP flux. Moreover, TIGAR can increase exercise endurance through improved mitochondrial functions via SIRT1/PGC-1 α [15].

Mitochondria are the principal organelle where the body produces energy, and its function and content directly affect the supply of ATP [16]. Moreover, fatigue is closely related to maintaining the balance between nutrition and energy metabolism [17]. Therefore, exercise-induced fatigue could be effectively alleviated by improving the mitochondrial biogenesis. Peroxisome proliferator-activated receptor-gamma coactivator one alpha (PGC-1 α) promotes mitochondrial biogenesis by activating transcription factors nuclear respirator factor1 (NRF1), which regulate mitochondrial transcription factor A (TFAM) [18]. TFAM can translocate to the mitochondrial matrix, and then stimulate mitochondrial DNA replication and mitochondrial gene expression [19]. Silent mating type information regulation 2 homolog-1 (Sirt1), an NAD⁺-dependent class III protein deacetylase belong to sirtuins family, is heavily implicated in health span and longevity [20]. A large number of studies have shown SIRT1 increases transcriptional activity of PGC-1 α to enhance mitochondrial biogenesis [21,22]. The previous study found that porcine whole-blood protein hydrolysate can enhance exercise performance and resistance to exercise-induced fatigue in mice by promoting mitochondrial biogenesis via SIRT1/PGC-1 α /NRF-1/TFAM signaling pathway [23].

Cordyceps militaris, a highly precious mushroom, is consisted of abundant natural active components with various biological activities and nutritional value [24]. It has been widely used as traditional medicine to maintain health and treat several diseases because of its powerfully therapeutic values for hundreds of years [25]. Cordycepin(3'-deoxyadenosine), a principal active ingredient, is a natural product extracted from *Cordyceps militaris* [26]. Previous studies have demonstrated that it has a lot of pharmacological activities, such as anti-diabetes [27], anti-hyperlipidemia [28], anti-inflammatory [29], immunomodulatory [30], antioxidant [31] effects. The previous study has shown that *Cordyceps militaris* can relieve fatigue [32], however, the active compound and specific mechanism have not been elucidated. This study aimed to investigate the beneficial effect of cordycepin on physical fatigue and illustrate the potential mechanism.

Section snippets

Materials and reagent

Cordycepin was provided by Jiangsu Synergetic Innovation Center for Advanced Bio-Manufacture (Nanjing, China). Lactic acid (LA), blood urea nitrogen (BUN), hepatic glycogen (HG), muscle glycogen (MG), glutathione (GSH), superoxide dismutase (SOD), malondialdehyde (MDA), nicotinamide adenine dinucleotide phosphate (NADPH) assay kits were purchased from Jiancheng Bioengineering Institute (Nanjing, China). BCA protein assay kit were purchased from (Bio-Rad, Hercules, CA, USA). The antibodies of ...

Effect of cordycepin on the body weight of mice

During 28 days of treatment this experiment, the mean body weight of mice in all groups presented a gradually incremental trend as the experimental process. However, there were no significant difference of body weight in all groups (Fig. 1B). ...

Effect of cordycepin on weight-loaded swimming time of mice

On the 28th day, weight-loaded swimming test was performed to assess the effect of cordycepin on exercise tolerance. As shown in Figs. 1C and 20 mg/kg and 40 mg/kg of cordycepin significantly increased the swimming time of weight-loaded swimming mice. The ...

Discussion

Cordyceps militaris, a traditional Chinese medicine with multiple active ingredients, have great pharmacological effects in health care and several diseases for hundreds of years [24,25]. Cordycepin, as an active compound of *Cordyceps militaris* [39], presents outstanding anti-inflammatory [40], antitumor [41], antioxidant [42], immunomodulatory [43], and neuroprotective [44] activities. Herein we presented the first report the anti-fatigue effect of cordycepin on the weight-loaded swimming mice ...

Conclusions

In summary, the current study proved the anti-fatigue effect of cordycepin on weight-loaded swimming mice by activating TIGAR/SIRT1/PGC-1 α signaling pathway. Our findings indicated that cordycepin might be a promising and beneficial anti-fatigue dietary supplements. ...

Author contributions

Yang Sun: Conceptualization, Methodology, Writing-Reviewing. Qi Zhang: Conceptualization, Supervision. Xiaoming Chai: Investigation, Data curation, Writing-Original draft preparation. Mengyue Pan: Software, Investigation. Jingjie Wang and Mingmei Feng: Investigation. Yupeng Wang: Software, Validation. ...

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. ...

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